

Attention-aware task scheduling under partial observability of human cognitive state: a memory-augmented deep reinforcement learning approach

P. N. Bhargav Teja^a, Lanka Sree Chathurya^a, Rajay Vedaraj I.S.^{a,*}

^aVellore Institute of Technology, Vellore, 632014, Tamil Nadu, India

Abstract

Classical dispatching rules such as earliest-deadline-first (EDF) and shortest-processing-time (SPT) assume that the resource executing work has constant capacity. This assumption fails when the resource is a person whose attention is depleted by sustained effort, whose fatigue accumulates, and whose alertness follows a circadian rhythm. This paper formulates human-centric task scheduling as a partially observable Markov decision process. Rest is an explicit action, attention and fatigue are latent states observed only through noisy estimates, and a multi-objective reward balances deadline-weighted task value against cognitive load. We propose an attention-aware deep Q-network (AA-DQN) that combines a short observation history, which approximates the belief state, with Double Q-learning, a dueling architecture and a hybrid action space of workload-intensity choices and dispatching rules. In a cognitively grounded simulator with stochastic task arrivals, AA-DQN was benchmarked against nine rule-based schedulers, including grid-tuned EDF and SPT variants that rest periodically or when observed fatigue or attention crosses a threshold, using ten training seeds, 500 held-out workdays with common random numbers, and multiplicity-corrected paired tests.

*Corresponding author.

Email address: rajay@vit.ac.in (Rajay Vedaraj I.S.)

Rules that never rest performed poorly (EDF: 46.8% of tasks on time). AA-DQN matched the return of the best tuned rule (difference -0.13 , 95% CI $[-0.29, 0.02]$) and achieved the highest on-time rate (72.8% versus 71.4%). Unlike any single rule, it combined this with high difficulty-weighted completion, avoiding the neglect of demanding tasks that afflicts SPT-based rules. The learned policy works intensively while the worker is fresh, concentrates breaks around the post-lunch circadian dip, and schedules demanding tasks when attention is high. Ablations and zero-shot tests identify when learning adds most value, namely with reliable cognitive-state estimates and with fatigue-prone workers, and where it remains limited.

7 *Keywords:* Human-centric scheduling, Deep reinforcement learning, Partially
8 observable Markov decision process, Cognitive fatigue, Attention, Rest-break
9 scheduling, Industry 5.0

10 **1. Introduction**

11 Scheduling decides which piece of work is executed when, and it is one of the
12 oldest problems in operations research and computer engineering [1–4]. Classical
13 dispatching rules such as First-In-First-Out (FIFO), Shortest Processing Time
14 (SPT) and Earliest Deadline First (EDF) are attractive because they are transparent,
15 cheap to compute and, under the right assumptions, provably optimal: EDF
16 minimises maximum lateness on a single resource with constant capacity [2, 5],
17 and SPT minimises mean flow time [1]. The common assumption underlying these
18 guarantees is that the resource executing the work has a *fixed, state-independent*
19 *processing capacity*. This assumption is reasonable for processors and machines
20 but is violated whenever the resource is a person.

21 Human work capacity is dynamic. Sustained effort depletes attentional re-
22 sources and produces the vigilance decrement first documented by Mackworth [6]
23 and since replicated across many tasks [7–9]; mental fatigue accumulates with
24 time-on-task and degrades both speed and accuracy [10–13]; and alertness follows
25 a circadian rhythm with a pronounced early-afternoon dip [14–17]. Rest breaks
26 partially reverse these effects, and field studies consistently report that well-timed
27 breaks preserve or even increase productivity despite the time they consume [18–
28 22]. A schedule that is optimal for a constant-capacity machine can therefore be
29 poor for a human: it front-loads demanding work regardless of cognitive state,
30 never allocates recovery, and drives the worker into a low-capacity regime in
31 which deadlines are missed precisely because capacity was ignored. The concern
32 is not only productivity. Chronic fatigue is associated with errors and accidents
33 [23, 24], and the move from Industry 4.0 towards a human-centric Industry 5.0
34 explicitly calls for production and service systems that adapt to the well-being of
35 their workers rather than the reverse [25–29].

36 Operations-management research has long recognised this gap. Models of
37 worker fatigue and recovery have been integrated into dual-resource-constrained
38 systems [30–32], rest-allowance and break-scheduling problems have been formu-
39 lated analytically [33–35], and taxonomies for merging scheduling theory with
40 human factors have been proposed [36–39]. These approaches, however, usually
41 assume that the worker’s state is known or deterministic and optimise an offline
42 plan; they do not provide an adaptive *policy* that reacts to the evolving, uncertain
43 state of a specific person during the day. Deep reinforcement learning (DRL),
44 in turn, has produced effective adaptive schedulers for job shops, clusters and
45 production lines [40–46], and recent surveys document rapid growth of the field

46 [47–49]. Yet the resources in these studies are machines, and the few recent studies
47 that include worker fatigue embed it in large shop-floor models where fatigue is a
48 deterministic, fully observed variable [50, 51]. Two features of the human problem
49 are thus rarely addressed together: *the decision of when to rest is itself a scheduling*
50 *action, and the cognitive state that should drive that decision cannot be measured*
51 *directly*. In practice, attention and fatigue are inferred from noisy proxies such
52 as physiological signals, eye tracking, performance probes or self-report [52–56],
53 which makes the scheduling problem partially observable.

54 This paper studies attention-aware task scheduling for a single knowledge or
55 production worker as a partially observable Markov decision process (POMDP)
56 [57–59] and solves it with a memory-augmented deep Q-network. We pay par-
57 ticular attention to the methodological issues that make DRL comparisons hard
58 to trust [60, 61]: we compare against strong, *tuned*, cognition-aware heuristics
59 rather than only against FIFO and EDF; we use disjoint seed sets for training,
60 tuning and testing; we evaluate on 500 held-out workdays with common random
61 numbers; we report ten independent training seeds with confidence intervals and
62 multiplicity-corrected paired tests; and we separate effects of the reward function
63 from effects of the dynamics through ablation.

64 The contributions are as follows.

- 65 (i) **A cognitively grounded scheduling simulator.** We formulate a workday-
66 scale scheduling environment in which a latent homeostatic attention re-
67 source, an exponentially accumulating and recovering fatigue state [30],
68 and a circadian process [14, 15] jointly determine the worker’s instanta-
69 neous work rate. Tasks arrive stochastically with heterogeneous difficulty,
70 work content and deadlines. Rest is an explicit action. The simulator, its

71 parameters and their behavioural rationale are fully specified and released.

72 (ii) **A POMDP formulation with a multi-objective reward.** The scheduler ob-
73 serves only noisy estimates of attention and fatigue together with a compact,
74 size-invariant summary of the task queue. The reward combines deadline-
75 weighted task value with explicit attention and fatigue terms, and we analyse
76 the resulting productivity–fatigue trade-off as a Pareto front [62, 63].

77 (iii) **An attention-aware DQN (AA-DQN).** We combine a finite observation
78 history, which acts as an approximate sufficient statistic for the belief state
79 [64–66], with Double Q-learning [67] and a dueling architecture [68]. A
80 component ablation quantifies the contribution of each element.

81 (iv) **A rigorous empirical study.** Against nine rule-based baselines, including
82 grid-tuned heuristics that combine EDF or SPT sequencing with periodic
83 or state-triggered rest and observe the same noisy cognitive signals, we test
84 (a) whether learning helps, (b) *why* it helps, through environment ablations
85 that remove observation noise, the circadian process, the cognitive dynamics
86 or the cognitive reward terms, (c) what behaviour is learned, and (d) how
87 the learned policy generalises to unseen worker profiles, workloads and
88 sensor-noise levels.

89 The remainder of the paper is organised as follows. Section 2 reviews related
90 work and positions the study. Section 3 formulates the problem. Section 4 presents
91 AA-DQN. Section 5 details the experimental protocol. Section 6 reports results,
92 Section 7 discusses implications, ethics and limitations, and Section 8 concludes.

93 2. Related work

94 2.1. Classical and dynamic scheduling

95 Single-resource sequencing theory provides the canonical dispatching rules
96 and their optimality conditions: SPT for mean flow time [1], EDF for maximum
97 lateness [2], and the Moore–Hodgson algorithm for the number of late jobs [69].
98 The three-field classification of Graham et al. [3] and the textbook treatment of
99 Pinedo [4] organise the deterministic theory. In real-time computing, EDF is
100 optimal for preemptive scheduling of independent tasks whenever the processor
101 is not overloaded [5], but its performance can collapse under overload, when
102 a domino effect causes many deadlines to be missed [70]. This failure mode is
103 directly relevant here, because a fatigued worker is in effect an overloaded resource.
104 Surveys of priority rules [71, 72] and of dynamic and reactive scheduling [73, 74]
105 show that practice relies heavily on such rules when information arrives online. In
106 all of these models, however, processing times are exogenous: they do not depend
107 on the history of the schedule.

108 2.2. Human factors in scheduling and workforce models

109 Personnel scheduling and rostering [75–77] assign workers to shifts but typ-
110 ically treat performance within a shift as constant. Research at the interface of
111 operations and human resources argues that this abstraction misses first-order
112 effects [37, 38], and Lodree et al. [36] proposed a taxonomy for integrating hu-
113 man factors into scheduling theory. Quantitative models of learning, forgetting,
114 fatigue and recovery have been developed for dual-resource-constrained systems
115 [30–32, 78]. They use exponential fatigue accumulation during work and expo-
116 nential recovery during rest, which is the functional form we adopt. Rest-break

117 scheduling has been formulated as an optimisation problem since Bechtold et
118 al. [33], with ergonomic guidance on work–rest ratios summarised by Konz [34]
119 and rest-allowance models for task assignment developed by Calzavara et al. [35].
120 Fatigue has also been incorporated into flexible job-shop formulations solved by
121 metaheuristics [50] and, very recently, by graph-based DRL [51]. Reviews of hu-
122 man factors in Industry 4.0 and in order picking [39, 79–81] conclude that worker
123 heterogeneity and well-being are still under-represented in operational models. A
124 common feature of this literature is that the worker state is assumed to be known,
125 either deterministic or computed from the schedule, so the decision problem is
126 fully observable.

127 *2.3. Deep reinforcement learning for scheduling*

128 Reinforcement learning has been applied to scheduling since the mid-1990s
129 [82, 83], and the combination of Q-learning [84] with deep function approximation
130 [85] enabled scheduling policies that operate on rich state representations. Notable
131 examples include cluster resource management [40], data-processing job schedul-
132 ing with graph neural networks [41], semiconductor production scheduling [42],
133 dynamic flexible job shops with job insertions [43, 46], and size-agnostic job-shop
134 dispatching [44, 45, 86]. Parameter-adaptive RL for dynamic job shops [87] and
135 RL for combinatorial optimisation more broadly [88, 89] complete the picture.
136 Systematic reviews [47–49] note three recurring weaknesses: comparisons against
137 untuned or weak baselines, limited statistical analysis, and little attention to human
138 resources. Our study addresses all three.

139 2.4. *Cognitive state, its measurement, and adaptive systems*

140 Resource theories of attention [90, 91] and cognitive load theory [92] model
141 mental capacity as a limited resource. Vigilance research shows that this resource
142 is depleted by sustained effort [6, 7] and partially restored by brief breaks [93].
143 Accounts of fatigue as an effort-regulation or opportunity-cost signal [12, 94]
144 and evidence linking fatigue to disengagement [95] motivate treating fatigue as a
145 state variable that shapes future capacity. Biomathematical fatigue models, from
146 the two-process model of sleep regulation [96–98] to the three-process model of
147 alertness [14] and applied models such as SAFTE [99], combine homeostatic and
148 circadian components. We borrow this decomposition at the time scale of a single
149 workday. Recovery research in organisational psychology confirms the value of
150 micro-breaks [22, 100–102]. On the measurement side, workload and fatigue can
151 be estimated from EEG, cardiac, ocular and behavioural signals [52–54, 103, 104],
152 from vigilance probes such as the PVT [55, 105], or from questionnaires such as
153 NASA-TLX [56, 106]. All of these estimates are noisy. Adaptive automation
154 and physiological computing use such estimates to adjust system behaviour [107–
155 112], and interruption-management systems decide when to notify a user based
156 on inferred cognitive state [113–116]. Reinforcement learning has been used
157 to personalise just-in-time health interventions [117–120], and in human–robot
158 collaboration to adapt task allocation to human capability and fatigue [121–125].
159 These works show that cognitive-state estimates can drive decisions, but they rarely
160 address the sequential scheduling of a task queue under deadlines.

161 2.5. *Partial observability and multiple objectives in RL*

162 When the state is not directly observed, the optimal policy is a function of the
163 belief state [57–59]. Memoryless policies can be arbitrarily poor [64]. Practical

164 deep RL approximates the belief either with recurrent networks [65, 126, 127]
 165 or with finite observation histories, as in the frame stacking of Mnih et al. [85].
 166 Ni et al. [66] show that well-tuned memory-based model-free agents are strong
 167 POMDP baselines. Stabilisation techniques for DQN, namely experience replay
 168 [128], target networks [85], Double Q-learning [67], dueling heads [68] and their
 169 combinations [129], are well established, as is the theory of TD learning with func-
 170 tion approximation [130]. Scheduling for humans is inherently multi-objective:
 171 productivity, deadlines and well-being conflict. Multi-objective RL provides the
 172 vocabulary of scalarisation and Pareto fronts [62, 63, 131]. Reward shaping must
 173 be used with care, because additional reward terms can change the optimal policy
 174 [132]. Finally, the deployment of RL in settings that involve people raises safety
 175 and reliability concerns [133–135], which we address in Section 7.

176 *2.6. Research gap*

177 Table 1 positions this work against representative streams. To our knowledge,
 178 no prior study simultaneously (i) models dynamic attention, fatigue and circadian
 179 variation as endogenous state, (ii) treats rest as a scheduling action, (iii) assumes
 180 that the cognitive state is only partially observable, (iv) learns an adaptive policy,
 181 and (v) benchmarks that policy against tuned, cognition-aware heuristics with
 182 multi-seed statistical testing. Our study fills this gap.

183 **3. Problem formulation**

184 We consider a single worker who processes a stream of tasks during one work-
 185 day. A scheduler, which could be a digital assistant, a workflow-management
 186 system or a supervisory controller in a human–robot cell, decides at regular inter-
 187 vals what the worker should do next: start or continue an easy task, a hard task, or

Table 1: Positioning of this study relative to representative research streams (✓: addressed; ◦: partially or in a simplified form; –: not addressed).

Stream (representative works)	Dynamic human state	Rest as action	Partial observ.	Adaptive policy	Deadlines/ queue	Tuned cog. baselines
Dispatching rules [1, 2, 5, 72]	–	–	–	–	✓	–
Fatigue/recovery models in DRC systems [30–32]	✓	◦	–	–	◦	–
Rest-break / rest-allowance optimisation [33–35]	◦	✓	–	–	–	–
DRL for machine scheduling [40, 42–44, 46]	–	–	–	✓	✓	–
Fatigue-aware shop schedul- ing [50, 51]	✓	–	–	◦	✓	–
Adaptive automation / inter- ruption mgmt. [107, 112– 114]	✓	◦	✓	◦	–	–
This study	✓	✓	✓	✓	✓	✓

188 take a short break. The worker’s capacity depends on a latent cognitive state that
189 evolves with the schedule. Table 2 summarises the notation and Table 3 lists the
190 parameter values.

191 3.1. Workload model

192 The day is divided into $T = 48$ slots of $\Delta = 10$ minutes (09:00–17:00). Each
193 day contains $N = 12$ tasks. Six tasks are available at the start of the day and
194 the remaining six arrive at slots drawn uniformly from $\{1, \dots, \lfloor 0.6T \rfloor\}$, which

Table 2: Notation.

Symbol	Meaning
$t \in \{0, \dots, T-1\}, \Delta$	decision slot index and slot length ($T = 48, \Delta = 10$ min)
h_t	clock time of slot t (h), $h_t = 9 + t\Delta/60$
a_i, d_i, w_i, D_i	arrival slot, difficulty, work content and deadline of task i
H_t	homeostatic attention resource (latent)
$c(h), A_t$	circadian modulation and effective attention, $A_t = \text{clip}(H_t + \gamma c(h_t))$
F_t	fatigue (latent)
ρ_t	work rate (work units completed per slot)
$\tilde{A}_t, \tilde{F}_t$	noisy observations of A_t, F_t
o_t, x_t	observation vector and history window of length k
u_t, b_t	indicators of working and resting in slot t
r_t	scalar reward; λ, μ fatigue and attention weights
γ_{RL}	RL discount factor (distinct from circadian amplitude γ)

195 models interruptions and new requests. Task i has difficulty $d_i \sim \mathcal{U}[0.2, 1.0]$ and
 196 is labelled *hard* if $d_i \geq 0.6$ and *easy* otherwise. Its work content is $w_i = 1 + 2.5 d_i$
 197 work units, so a task requires between 1.5 and 3.5 slots (15–35 min) at full capacity.
 198 Its deadline is $D_i = \min\{T, a_i + \lceil s_i T \rceil\}$ with relative slack $s_i \sim \mathcal{U}[0.3, 1.0]$. With
 199 these settings, the expected daily work content (≈ 30 units) is below the capacity
 200 of a fully rested worker (48 units) but above what a worker can deliver once
 201 attention and fatigue degrade. Scheduling decisions therefore matter: the day is
 202 *conditionally overloaded*.

203 3.2. Cognitive dynamics

204 *Effective attention.* Following the decomposition of alertness into homeostatic and
 205 circadian components [14, 96, 98], effective attention is

$$A_t = \text{clip}(H_t + \gamma c(h_t), 0, 1), \quad c(h) = 0.2 e^{-(h-10.5)^2/(2 \cdot 1.5^2)} - e^{-(h-14)^2/(2 \cdot 1.2^2)}, \quad (1)$$

206 where $c(h)$ encodes a late-morning peak and the post-lunch dip around 14:00
 207 [15–17] (Fig. 1a), and $\gamma = 0.15$ sets its amplitude.

208 *Homeostatic attention resource.* Working on a task of difficulty d depletes the
 209 attentional resource, and the depletion is accelerated by fatigue. This captures the
 210 vigilance decrement and resource-control accounts of sustained attention [6–8].
 211 Resting restores the resource exponentially towards its maximum [22, 93]:

$$H_{t+1} = \text{clip}\left(H_t - \alpha d (1 + \phi F_t) u_t + \beta (1 - H_t) b_t + \sigma_p \xi_t, 0, 1\right), \quad \xi_t \sim \mathcal{N}(0, 1), \quad (2)$$

212 where $u_t, b_t \in \{0, 1\}$ indicate work and rest ($u_t + b_t = 1$), $\alpha = 0.02$ is the depletion
 213 rate per unit difficulty, $\phi = 0.5$ the fatigue coupling, $\beta = 0.30$ the per-slot recovery
 214 fraction, and $\sigma_p = 0.02$ the process-noise scale that represents unmodelled day-
 215 to-day and moment-to-moment variability.

216 *Fatigue.* Fatigue accumulates exponentially towards saturation during work and
 217 decays exponentially during rest. This is the discrete-time analogue of the fatigue-
 218 recovery model of Jaber and Neumann [30], also used in [31, 32]:

$$F_{t+1} = \text{clip}\left(F_t + \delta d (1 - F_t) u_t - \eta F_t b_t, 0, 1\right), \quad (3)$$

219 with accumulation rate $\delta = 0.05$ and recovery rate $\eta = 0.20$.

220 *Productivity.* While working on task i , the worker completes

$$\rho_t = \rho_0 A_t^{0.5+d_i} (1 - \kappa F_t) \quad (4)$$

221 work units in slot t , with $\rho_0 = 1$ and $\kappa = 0.5$. The exponent makes hard tasks more
 222 sensitive to attention than easy ones (Fig. S1 in the Supplementary Material), in
 223 line with resource theories in which demanding tasks draw more heavily on limited
 224 capacity [90, 91]. The fatigue factor reduces throughput multiplicatively [10, 12].
 225 A task is complete at the end of the first slot in which its cumulative work reaches
 226 w_i .

227 *Initial state and behavioural plausibility.* Each day starts with $H_0 \sim \mathcal{U}[0.8, 1.0]$
 228 and $F_0 \sim \mathcal{U}[0, 0.15]$. Parameters were chosen *a priori* to reproduce qualitative
 229 patterns reported in the time-on-task and fatigue literature rather than fitted to a
 230 particular dataset. Under continuous work without breaks, attention falls from
 231 ≈ 1.0 to ≈ 0.3 by the end of the day, with a visible early-afternoon trough, and
 232 fatigue rises to ≈ 0.85 . One 10-minute break recovers 30% of the attentional deficit
 233 and 20% of accumulated fatigue (Fig. 1b,c). Because no single parameterisation
 234 can represent all workers, Section ?? evaluates the policies zero-shot on resilient
 235 and fatigue-prone profiles in which all four rate parameters are scaled by $\pm 30\%$.

236 3.3. Observation model and partial observability

237 The scheduler cannot observe H_t , A_t or F_t . It receives noisy estimates

$$\tilde{A}_t = \text{clip}(A_t + \sigma_o \epsilon_t^A, 0, 1), \quad \tilde{F}_t = \text{clip}(F_t + \sigma_o \epsilon_t^F, 0, 1), \quad \epsilon_t^A, \epsilon_t^F \sim \mathcal{N}(0, 1), \quad (5)$$

238 with $\sigma_o = 0.10$ in the nominal setting. These estimates stand in for wearable, oc-
 239 ular, behavioural or self-report estimators, whose errors are substantial in practice

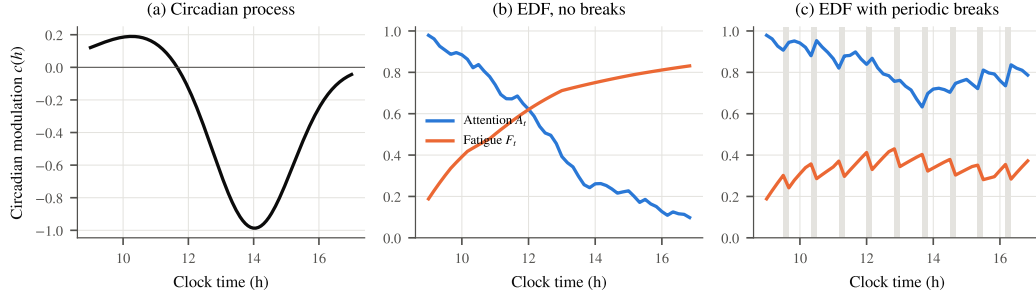


Figure 1: Simulator dynamics. (a) Circadian modulation $c(h)$ with a late-morning peak and post-lunch dip. (b) Effective attention and fatigue on one held-out test day under EDF without breaks. (c) The same day under EDF with a break after every four work slots (shaded bands mark breaks). Breaks keep attention high and fatigue low at the cost of working time.

240 [52–54]. The full observation is the 13-dimensional vector

$$o_t = \left[\frac{t}{T}, \tilde{A}_t, \tilde{F}_t, \underbrace{\frac{n_t^e}{N}, \frac{W_t^e}{20}, \sigma_t^e, \frac{m_t^e}{N}, \frac{\omega_t^e}{4}}_{\text{easy class}}, \underbrace{\frac{n_t^h}{N}, \frac{W_t^h}{20}, \sigma_t^h, \frac{m_t^h}{N}, \frac{\omega_t^h}{4}}_{\text{hard class}} \right], \quad (6)$$

241 where, for each class, n_t is the number of available unfinished tasks, W_t their
 242 remaining work, $\sigma_t = \text{clip}((\min_i D_i - t)/T, -1, 1)$ the normalised slack of the
 243 most urgent task (1 if the class is empty), m_t the number of overdue tasks, and ω_t
 244 the remaining work of the shortest task (0 if the class is empty). Unlike a binary
 245 completion vector, this summary is invariant to the number of tasks and exposes
 246 the deadline information that urgency-aware scheduling requires. The worker’s
 247 true state and the future arrivals are hidden, so the decision problem is a POMDP
 248 $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma_{\text{RL}} \rangle$ [59]. Here $\mathcal{S}$ contains the full task set and (H_t, F_t, t) , Ω
 249 is the observation space of Eq. (6), O is given by Eq. (5), and P by Eqs. (1)–(4)
 250 together with the arrival process.

251 3.4. Action space

252 The agent chooses among five macro-actions,

$$\mathcal{A} = \{\text{EASY}, \text{HARD}, \text{EDF}, \text{SPT}, \text{BREAK}\}. \quad (7)$$

253 EASY (HARD) dispatches the available easy (hard) task with the earliest deadline,
 254 falling back to the other class if the chosen class is empty. EDF and SPT dispatch,
 255 across both classes, the task with the earliest deadline or the least remaining work.
 256 BREAK allocates the slot to rest, and the worker also rests when no task is available.
 257 The first two actions let the agent regulate the cognitive *intensity* of work relative
 258 to the worker’s state. The two rule actions let it switch between urgency-driven and
 259 throughput-driven sequencing, in the spirit of DRL schedulers that select among
 260 dispatching rules [43, 48]. This hybrid design has an important methodological
 261 consequence: every rule-based baseline in Section 5.1 can be expressed as a
 262 policy over $\mathcal{A}$, so a learned policy is never handicapped relative to the baselines
 263 by its action space. Section ?? reports an agent restricted to the intensity actions
 264 $\{\text{EASY}, \text{HARD}, \text{BREAK}\}$ of the original formulation. The action space is independent
 265 of N , and tasks are preemptible at slot boundaries, which is common in knowledge
 266 work.

267 3.5. Reward

268 The per-slot reward combines task outcomes with cognitive terms:

$$r_t = \underbrace{\sum_{i \in C_t} (1 + d_i) [\mathbb{I}(t + 1 \leq D_i) + 0.25 \mathbb{I}(t + 1 > D_i)] - 0.5 |\mathcal{M}_t| + \mu A_{t+1} - \lambda F_{t+1}}_{r_t^{\text{task}}}, \quad (8)$$

269 where C_t is the set of tasks completed in slot t , $\mathcal{M}_t$ is the set of tasks whose deadline
 270 expires unfinished at the end of slot t , $\mu = 0.05$ and $\lambda = 0.10$. Harder tasks are
 271 worth more, late completion retains a quarter of the value, and each missed deadline
 272 costs 0.5. The attention and fatigue terms express the organisation’s preference for
 273 sustainable work. Because such terms can change the optimal policy [132], we (i)
 274 report reward-independent operational metrics (on-time rate, completion, fatigue
 275 exposure) alongside the return, (ii) train an agent with $\lambda = \mu = 0$ to show that
 276 the learned behaviour is driven by the dynamics rather than by the reward terms,
 277 and (iii) sweep λ to trace the productivity–fatigue Pareto front. The objective
 278 is to find a policy π that maps observation histories to actions and maximises
 279 $\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma_{\text{RL}}^t r_t \right]$.

280 **4. Attention-aware deep Q-network**

281 Figure 2 gives an overview of the closed loop between the simulated workday
 282 and the learning agent.

283 *4.1. From Q-learning to deep Q-networks*

284 For a fully observable MDP, the optimal action-value function satisfies the Bell-
 285 man optimality equation $Q^*(s, a) = \mathbb{E}[r + \gamma_{\text{RL}} \max_{a'} Q^*(s', a') \mid s, a]$ [137, 138],
 286 and tabular Q-learning converges to Q^* under standard conditions [84]. With con-
 287 tinuous cognitive variables, a tabular representation is infeasible, so Q is approx-
 288 imated by a neural network $Q(\cdot, \cdot; \theta)$ trained to minimise the temporal-difference
 289 (TD) error. The DQN of Mnih et al. [85] stabilises this nonlinear approximation
 290 with an experience replay buffer [128], which breaks the temporal correlation
 291 of samples, and a periodically synchronised target network $\bar{\theta}$. Convergence is

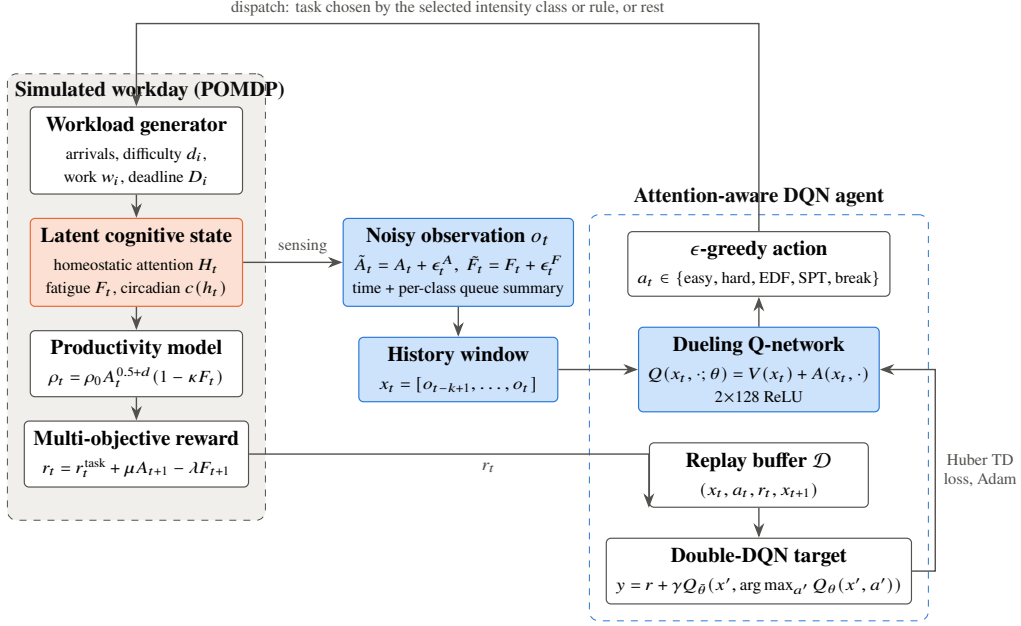


Figure 2: Overview of the attention-aware scheduling framework. The worker’s cognitive state is latent; the agent receives noisy estimates together with a queue summary, encodes a short observation history, and selects an intensity class, a dispatching rule or a break. Transitions are stored in a replay buffer and used to train a dueling Q-network with Double-DQN targets.

not guaranteed for nonlinear function approximation [130], so we assess stability empirically across independent seeds.

4.2. Handling partial observability with an observation history

In a POMDP, the optimal policy is a function of the belief $b_t(s) = \Pr(s_t = s \mid o_{0:t}, a_{0:t-1})$ [57, 59]. Exact belief tracking is intractable for our continuous, nonlinear dynamics, and a memoryless policy $\pi(o_t)$ can be arbitrarily suboptimal [64]. Two structural properties of the scheduling problem make a short history informative. First, the latent state is persistent. H_t and F_t change slowly relative to the observation noise, so averaging several consecutive estimates reduces the

error: for k independent observations of a quasi-static quantity, the standard error falls by a factor $\sqrt{k}$. Second, the trajectory of the observations, for example fatigue rising after consecutive hard tasks or attention recovering after a break, reveals the direction and rate of change, which a single noisy reading cannot. We therefore feed the network the window

$$x_t = [o_{t-k+1}, \dots, o_t] \in \mathbb{R}^{13k}, \quad k = 4, \quad (9)$$

zero-padded at the start of the day. This is the finite-memory approximation used for Atari [85]. It is computationally cheaper than recurrent alternatives [65, 127] and is a strong baseline for many POMDPs [66]. A 40-minute window spans the time scale of a typical work bout. Section ?? varies $k \in \{1, 2, 4, 8\}$.

4.3. Double Q -learning and dueling architecture

Max-based targets overestimate action values in noisy environments [67], and our environment is noisy by design. AA-DQN therefore uses the Double-DQN target, which decouples action selection (online network) from evaluation (target network):

$$y_t = r_t + \gamma_{\text{RL}}(1 - \text{done}_t) Q(x_{t+1}, \arg \max_{a'} Q(x_{t+1}, a'; \theta); \bar{\theta}). \quad (10)$$

In many states, for example at the start of the day or when the queue is empty, the choice between easy and hard work matters much less than the value of the state itself. A dueling head [68] represents this explicitly:

$$Q(x, a; \theta) = V(x; \theta) + \left(A(x, a; \theta) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(x, a'; \theta) \right). \quad (11)$$

The shared body consists of two fully connected layers of 128 ReLU units. The loss is the Huber loss of the TD error, which is robust to the occasional large

320 reward of a hard task completed on time [139]:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,r,x') \sim \mathcal{D}} [\ell_{\delta=1}(Q(x,a;\theta) - y)]. \quad (12)$$

321 Parameters are updated with Adam [140] and gradient-norm clipping. The vanilla
 322 DQN baseline uses the same action set and observation but $k = 1$, the standard max
 323 target and a single linear output head. This isolates the contribution of memory
 324 and of the two stabilisation components.

325 4.4. Exploration and training procedure

326 Actions are selected ϵ -greedily with an exponentially decaying exploration
 327 rate $\epsilon_n = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})e^{-\kappa_\epsilon n}$, where n is the environment step, $\epsilon_{\max} = 1$
 328 and $\epsilon_{\min} = 0.01$, and κ_ϵ is set so that ϵ reaches 0.05 after half of the training
 329 budget. Algorithm 1 summarises training. Every 100 episodes, the greedy policy
 330 is evaluated on a fixed set of 100 *validation* days that are disjoint from the training
 331 and test days, and the parameters with the highest validation return are retained.
 332 This mirrors the tuning of the heuristic baselines on their own held-out days
 333 (Section 5.1), so that both families receive the same model-selection budget and
 334 neither sees the test days. For transparency we also report the final-iterate policy
 335 (Supplementary Table S3).

336 4.5. Computational complexity

337 A forward pass costs $O(13k \cdot 128 + 128^2 + 128 \cdot |\mathcal{A}|) \approx 2.4 \times 10^4$ multiply–
 338 accumulate operations, independent of the number of tasks, because the observa-
 339 tion summarises the queue per class. The dispatch step is $O(N \log N)$. Inference
 340 therefore takes microseconds on a CPU (Section ??), which is negligible relative to
 341 a 10-minute decision interval and allows deployment on wearable or edge devices.

Algorithm 1 Training of the attention-aware DQN (AA-DQN).

Require: episodes E , history length k , replay capacity $|\mathcal{D}|$, batch size B , target period C , validation days $\mathcal{V}$

- 1: initialise θ randomly, $\bar{\theta} \leftarrow \theta$, $\mathcal{D} \leftarrow \emptyset$, $n \leftarrow 0$, $J^* \leftarrow -\infty$
- 2: **for** episode $e = 1, \dots, E$ **do**
- 3: sample a workday (tasks, initial state, noise); observe o_0 ; $x_0 \leftarrow [\mathbf{0}, \dots, \mathbf{0}, o_0]$
- 4: **for** $t = 0, \dots, T - 1$ **do**
- 5: $a_t \leftarrow$ random action with prob. ϵ_n , else $\arg \max_a Q(x_t, a; \theta)$
- 6: dispatch a_t (EDF within class or rest); simulate Eqs. (1)–(4); receive r_t (Eq. (8))
and o_{t+1}
- 7: $x_{t+1} \leftarrow \text{shift}(x_t, o_{t+1})$; store $(x_t, a_t, r_t, x_{t+1}, \text{done}_t)$ in $\mathcal{D}$; $n \leftarrow n + 1$
- 8: **if** $|\mathcal{D}| \geq \text{warm-up}$ **then**
- 9: sample a minibatch of B transitions; compute targets with Eq. (10)
- 10: $\theta \leftarrow \theta - \eta_{lr} \text{Adam}(\nabla_{\theta} \mathcal{L}(\theta))$ with gradient clipping
- 11: **if** $n \bmod C = 0$ **then** $\bar{\theta} \leftarrow \theta$
- 12: **if** $e \bmod 100 = 0$ **then**
- 13: $J \leftarrow$ mean greedy return on $\mathcal{V}$; **if** $J > J^*$ **then** $J^* \leftarrow J$, $\theta^* \leftarrow \theta$
- 14: **return** θ^*

342 5. Experimental design

343 5.1. Baselines

344 We compare AA-DQN against the vanilla DQN and nine rule-based sched-
345 ulers, which span classical dispatching and hand-crafted cognition-aware policies.
346 Baselines dispatch individual tasks directly, so they have at least as much control
347 over task choice as the RL agents.

- 348 • **Random:** uniform choice among EASY, HARD and BREAK.

- 349 • **FIFO**: the available task that arrived first; never rests.
- 350 • **EDF**: the available task with the earliest deadline; never rests [2, 5].
- 351 • **SPT**: the available task with the least remaining work; never rests [1].
- 352 • **EDF-Periodic**: EDF with one break after every p work slots, a fixed work–
353 rest cycle in the spirit of ergonomic rest-allowance guidance [34].
- 354 • **EDF-Threshold**: EDF that rests whenever the *observed* fatigue exceeds τ_F
355 or the observed attention falls below τ_A . This is a reactive, state-aware rule.
- 356 • **SPT-Periodic** and **SPT-Threshold**: the same two rest rules combined with
357 SPT sequencing. SPT limits the damage of overload by finishing short
358 tasks first, so these variants combine the best classical sequencing rule (Sec-
359 tion 6.2) with state-aware rest. They were added after a pre-submission
360 review identified them as the strongest hand-designed competitors (Supple-
361 mentary Section S5).
- 362 • **Cog-Heuristic**: rests on the same threshold rule and otherwise schedules a
363 hard task only when observed attention is at least τ_H , an easy task otherwise
364 (EDF within class). This hand-crafted rule encodes the intuition that the
365 learned policy might discover.

366 The parameters of the five parametric heuristics were selected by exhaustive
367 grid search, maximising mean return on 300 *tuning* days that are disjoint from
368 the training, validation and test days. The grids were $p \in \{2, 3, 4, 5, 6, 8, 10\}$;
369 $\tau_F \in \{0.3, \dots, 0.9, 1.01\}$ and $\tau_A \in \{0, 0.2, \dots, 0.7\}$ (56 combinations, for both
370 threshold rules); and, for Cog-Heuristic, $\tau_F \times \tau_A \times \tau_H$ with $\tau_H \in \{0.4, \dots, 0.9\}$

371 (252 combinations). The heuristics observe exactly the same noisy signals as
 372 the agents. The selected parameters are reported in Supplementary Table S2.
 373 Tuning is repeated separately for every environment variant, so that each ablation
 374 compares the agent with the best heuristic *for that variant*.

375 5.2. Protocol

376 The main configurations were trained with ten independent seeds, the environ-
 377 ment ablations and the domain-randomised variant with five, and the reward-weight
 378 sweep and the component study with three, each for 5000 episodes (2.4×10^5 de-
 379 cisions). Four disjoint sets of days were used: training days (generated from the
 380 training seed), 100 validation days for checkpoint selection, 300 tuning days for
 381 heuristic parameters, and 500 test days used only once, for the final evaluation.
 382 All methods are evaluated greedily on the *same* 500 test days with *common ran-*
 383 *dom numbers*. Each test day fixes the task set, the initial cognitive state and the
 384 sequences of process and observation noise, so differences between methods are
 385 paired and not confounded by sampling variation. In total, 75 agents were trained
 386 and more than 10^5 test days were simulated.

387 5.3. Metrics

388 Besides the return (Eq. (8)), which depends on the reward weights, we re-
 389 port reward-independent operational metrics: the *on-time rate* (fraction of the
 390 day’s tasks completed by their deadline); the *difficulty-weighted on-time rate*
 391 $\sum_i d_i \mathbb{1}(\text{on time}) / \sum_i d_i$; the *completion rate* (completed at any time); the mean
 392 *true* attention $\bar{A}$ and fatigue $\bar{F}$ over the day; the *high-fatigue exposure*, i.e. the
 393 fraction of slots with $F_t > 0.7$; and the *break share*.

394 5.4. Statistical analysis

395 Following recommendations for reliable RL evaluation [60, 61], we treat the
396 training seed as the unit of replication for learned policies. For each seed, metrics
397 are averaged over the 500 test days. We report the mean and standard deviation
398 across seeds, and 95% bootstrap confidence intervals [141] with 5000 resamples.
399 Deterministic heuristics are summarised across test days. To compare AA-DQN
400 with each baseline, we average AA-DQN over seeds for each test day and apply the
401 two-sided Wilcoxon signed-rank test [142] to the 500 paired day-level differences.
402 We report the matched-pairs rank-biserial correlation r_{rb} as effect size, the win
403 rate, and Holm-adjusted p -values [143] over the nine baseline comparisons. As
404 a more conservative seed-level check, we also report how many of the ten seeds
405 outperform each baseline.

406 5.5. Implementation

407 The simulator, agents and baselines are implemented in Python with NumPy
408 and PyTorch [144]. Table 4 lists the hyperparameters. Training used one CPU
409 thread per run. The code, configuration files, raw per-day results and analysis
410 scripts are released (see Data availability) so that every number and figure in this
411 paper can be regenerated with a single command.

412 6. Results

413 6.1. Learning dynamics

414 Figure 3 shows greedy validation performance during training, averaged over
415 ten seeds. AA-DQN improves rapidly during the first 1000 episodes and slowly
416 thereafter; it is still improving at the end of the 5000-episode budget. The seed-
417 averaged curve of the *current* parameters remains below the best tuned heuristic

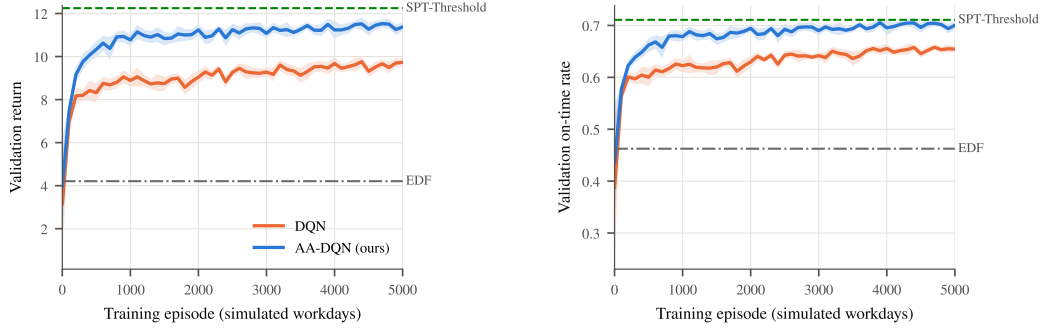


Figure 3: Greedy validation performance of the current parameters during training (mean of ten seeds; shaded bands are 95% confidence intervals). Horizontal lines show the best tuned heuristic (SPT-Threshold) and EDF on the same 100 validation days. Test results use the validation-selected checkpoint of each seed.

418 (SPT-Threshold) throughout training, in both return and on-time rate. This average,
 419 however, hides substantial variation between checkpoints within each run: the
 420 best checkpoints of every seed reach the heuristic’s level, while neighbouring
 421 checkpoints fall below it. Such policy oscillation is typical of value-based deep
 422 RL with a constant learning rate [60], and it motivates selecting checkpoints on
 423 held-out validation days. Validation-selected checkpoints were taken at episode
 424 3350 on average for AA-DQN and 3490 for DQN. Because validation and test days
 425 are disjoint, the test results below are not inflated by this selection. The vanilla
 426 DQN learns more slowly and plateaus well below AA-DQN. Its training return is
 427 also less stable (Supplementary Fig. S2), consistent with the over-estimation that
 428 Double Q-learning mitigates [67] and with the known weakness of memoryless
 429 policies in POMDPs [64].

430 6.2. Comparison with baselines

431 Table 5 and Figs. 4–5 report performance on the 500 held-out test days, and
432 Table 6 gives the paired statistical comparisons. Four findings stand out.

433 (i) *Capacity-agnostic rules fail under cognitive dynamics.* The rules that never
434 rest (FIFO, EDF, SPT) drive the worker into a low-capacity regime: mean fatigue
435 reaches 0.48–0.55, attention falls to 0.46–0.54, and EDF completes only 46.8% of
436 tasks on time. EDF performs particularly poorly, which reproduces the overload
437 domino effect known from real-time scheduling [70]: once capacity falls, chasing
438 the most urgent task means working on tasks that will be late anyway. SPT limits
439 the damage by finishing short tasks first (62.5%).

440 (ii) *State-aware rest is the dominant lever.* Adding rest to these rules, either
441 periodically or when observed fatigue or attention crosses a tuned threshold, raises
442 the return by 4–8 units and roughly halves mean fatigue. All competitive policies
443 rest in 15–20% of slots. The best rule overall is SPT-Threshold (return 12.30,
444 on-time rate 71.4%), which combines SPT sequencing with threshold-triggered
445 rest.

446 (iii) *AA-DQN matches the best tuned rule on return and is the only policy that is*
447 *strong on both completion measures.* AA-DQN achieves a return of 12.17 ± 0.15
448 across seeds. This is statistically indistinguishable from SPT-Threshold ($\Delta =$
449 -0.13 , 95% CI $[-0.29, 0.02]$, $p_{\text{Holm}} = 0.24$), and it significantly exceeds every
450 other baseline ($p_{\text{Holm}} < 10^{-4}$). It attains the highest on-time rate of all methods
451 (72.8%), significantly above SPT-Threshold (+1.4 percentage points $[0.7, 2.1]$,
452 $p_{\text{Holm}} = 0.00016$, all ten seeds better). The difficulty-weighted on-time rate reveals
453 why the two leading families differ (Fig. 5). SPT-based rules complete many tasks
454 on time by finishing short tasks first, but they neglect hard tasks (weighted on-time

455 rate 59.7%). EDF-based rules protect hard tasks (65.7%) but complete fewer tasks
456 overall. AA-DQN combines the strengths of both: it completes as many tasks on
457 time as any rule, while matching the EDF rules on difficulty-weighted completion
458 (65.5%; +5.7 pp over SPT-Threshold, [4.9, 6.6]). In other words, the learned
459 policy avoids the starvation of long tasks that is the classical weakness of SPT [4].
460 No single hand-designed rule in our set achieves this.

461 *(iv) The learned policy pays for this with somewhat more fatigue and fewer late*
462 *completions.* AA-DQN rests less than the tuned heuristics (15.0% versus 18.7% of
463 slots for SPT-Threshold). Its mean fatigue is therefore higher (0.333 versus 0.268),
464 although still 39% below EDF, and fatigue never exceeds 0.7. Because it focuses
465 effort on tasks that can still meet their deadline, it also completes fewer tasks
466 eventually (77.1% versus 81.7% for SPT-Threshold, counting late completions).
467 This explains why its return, which rewards late completions and low fatigue, is
468 not higher despite the higher on-time rate. Section ?? shows that the fatigue level
469 can be shifted by the reward weight.

470 Relative to the memoryless vanilla DQN, adding memory, Double Q-learning
471 and the dueling head increases the return by 1.88 ([1.80, 1.96]), with a win on
472 98.0% of test days. The vanilla DQN would rank only seventh among the eleven
473 methods, which shows that a naive application of DQN to this POMDP would
474 wrongly suggest that learning is not competitive. Validation-based checkpoint
475 selection matters for both agents (Supplementary Table S3): the final-iterate AA-
476 DQN has a return of 11.55 and an on-time rate of 71.0%, and the final-iterate DQN
477 has a return of 9.84. We report the validation-selected policies as primary results
478 because the heuristics received an equivalent selection step (grid search) on their
479 own held-out days.

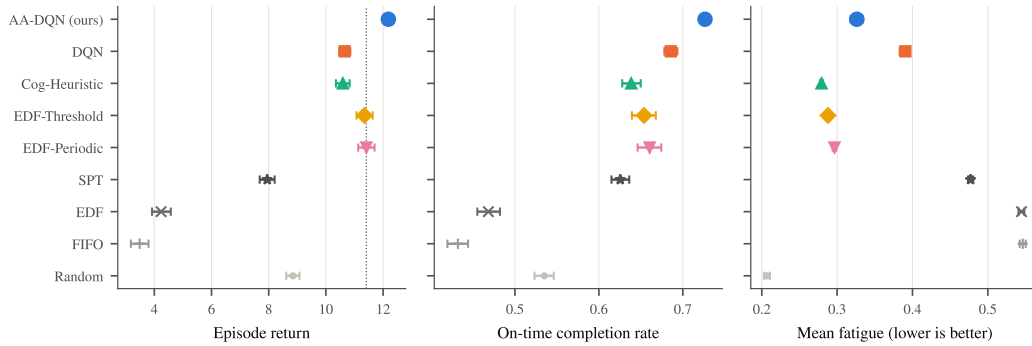


Figure 4: Test performance with 95% bootstrap confidence intervals (learned policies: across seeds; heuristics: across days). The dotted line marks the best heuristic return.

7. Discussion

7.1. Ethical and organisational considerations

A scheduler that monitors attention and fatigue and recommends when a person should work on what and when to rest touches on autonomy, privacy and fairness, and these issues must be designed in rather than added later [108, 112, 133]. We see four requirements for responsible deployment. First, *the system should be advisory*. Recommendations should be presented to the worker, who retains the authority to accept or override them; evidence from human–robot teaming indicates that the allocation of decision authority affects both efficiency and satisfaction [121]. Second, *cognitive-state data are sensitive personal data*. Physiological and behavioural signals should be processed with informed consent, ideally on the worker’s own device, and should never be used for performance appraisal. The observation model of AA-DQN needs only two scalar estimates per slot, which supports such data minimisation. Third, *fairness across workers* requires that policies do not systematically disadvantage people whose fatigue dynamics differ from the nominal profile. Section ?? shows how performance changes for resilient

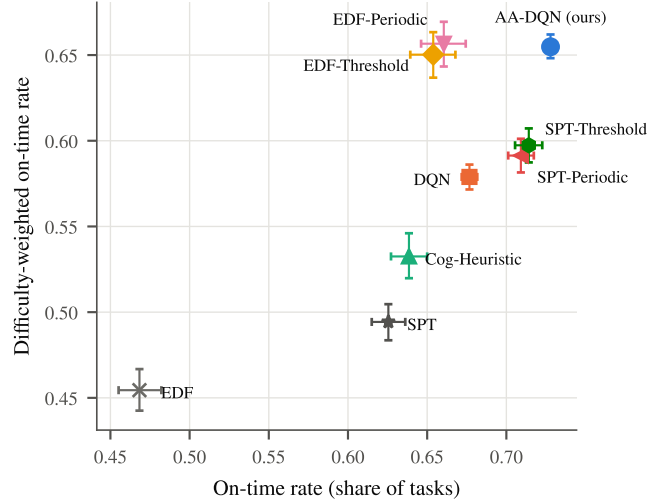


Figure 5: Count-based versus difficulty-weighted on-time completion (95% CIs). SPT-based rules complete many tasks but neglect hard ones; EDF-based rules protect hard tasks but complete fewer. AA-DQN is the only policy near the best value on both axes.

and fatigue-prone profiles, and personalisation (Section 7.3) would further reduce such disparities. Fourth, *safety constraints* should bound the policy, for example by guaranteeing a minimum amount of rest per period regardless of deadline pressure, consistent with safe RL principles [134]. Notably, the learned policies in this study already increase rest relative to deadline-driven rules. The reward weight λ makes the productivity–well-being trade-off explicit and auditable (Section ??), so stakeholders can choose it deliberately rather than leaving it implicit in a heuristic.

7.2. Limitations

The study has limitations that define its scope. (i) *Simulation, not human data*. The cognitive model is grounded in established functional forms and its parameters were chosen to be behaviourally plausible, but it has not been calibrated against individual time-series of attention and fatigue. The absolute numbers should

508 therefore be interpreted comparatively. The robustness analyses mitigate, but do
509 not remove, this concern. (ii) *Abstracted sensing*. Cognitive-state estimation is
510 modelled as additive Gaussian noise on the true state. Real estimators can be
511 biased, drift over the day, or fail intermittently, and they may be correlated with
512 task type. (iii) *A single worker and a single day*. Multi-day carry-over of fatigue,
513 sleep, and team-level interactions such as hand-offs and shared deadlines are
514 outside the scope. (iv) *Action abstraction*. The agent chooses a difficulty class and
515 delegates the choice of task within a class to EDF. This keeps the policy compact
516 and size-invariant but excludes policies that, for example, deliberately abandon a
517 hopeless task. (v) *Algorithmic scope*. We study value-based agents with a finite
518 history. Recurrent, model-based or policy-gradient agents [65, 127, 145] might
519 extract more from the observation sequence. (vi) *Simulated time-on-task effects*
520 *only*. Motivation, task interest, interruptions and errors, which also shape real
521 productivity [115, 116, 146], are not modelled.

522 7.3. Future work

523 Three extensions follow directly. First, *empirical calibration and validation*:
524 fitting the latent dynamics to longitudinal data from vigilance probes, wearables
525 and task logs [52, 55, 104], then evaluating the scheduler in a controlled user study
526 with NASA-TLX and performance outcomes [56], ideally as a micro-randomised
527 trial [117]. Second, *personalisation*: domain randomisation over worker profiles
528 during training [147, 148], and Bayesian or meta-learning adaptation to an individ-
529 ual’s parameters, in the spirit of personalised just-in-time interventions [118, 119].
530 Third, *multi-worker and human–robot settings*: extending the formulation to teams
531 and to collaborative cells in which a robot absorbs demanding sub-tasks when a
532 co-worker’s attention is low [122–125], with constrained or multi-objective RL to

533 enforce explicit well-being guarantees [63].

534 **8. Conclusion**

535 TBD

536 **CRedit authorship contribution statement**

537 **P.N. Bhargav Teja:** Conceptualization, Methodology, Software, Formal
538 analysis, Investigation, Visualization, Writing – original draft. **Lanka Sree**
539 **Chathurya:** Methodology, Software, Validation, Data curation, Investigation,
540 Writing – original draft. **Rajay Vedaraj I.S.:** Conceptualization, Supervision,
541 Methodology, Resources, Project administration, Writing – review & editing.

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543 The authors declare that they have no known competing financial interests or
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549 **Data availability**

550 The simulator, agents, baselines, experiment configurations, raw per-episode
551 results and the scripts that generate every table and figure are openly available at

552 <https://github.com/bhargavteja-9779/r1>. No human-subject data were
553 collected.

554 **Declaration of generative AI and AI-assisted technologies in the manuscript** 555 **preparation process**

556 During the preparation of this work the authors used an AI assistant (Claude,
557 Anthropic) to support code development, language editing and the organisation of
558 the manuscript. After using this tool, the authors reviewed and edited the content
559 as needed and take full responsibility for the content of the published article.

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Table 3: Nominal simulator parameters and their rationale.

Parameter	Value	Rationale / source
Slots per day T , slot length Δ	48, 10 min	8-h workday; decision granularity of micro-breaks [22]
Tasks per day N (initial / arriving)	12 (6 / 6)	conditionally overloaded day; stochastic arrivals
Difficulty d_i , hard threshold	$\mathcal{U}[0.2, 1]$, 0.6	range of the original formulation
Work content w_i	$1 + 2.5d_i$ units	15–35 min at full capacity
Relative slack s_i	$\mathcal{U}[0.3, 1]$	range of the original formulation
Attention depletion α , fatigue coupling ϕ	0.02, 0.5	vigilance decrement over hours of work [7, 8]
Attention recovery β	0.30 per slot	break benefits [22, 93]
Circadian amplitude γ	0.15	post-lunch dip [15, 16]
Fatigue accumulation δ , recovery η	0.05, 0.20	exponential fatigue–recovery [30, 31]
Fatigue throughput penalty κ	0.5	performance loss under fatigue [10, 136]
Process / observation noise σ_p, σ_o	0.02, 0.10	variability; sensor error [52, 54]
Reward weights μ, λ	0.05, 0.10	swept in Section ??

Table 4: Agent hyperparameters (shared by DQN and AA-DQN unless stated).

Hyperparameter	DQN	AA-DQN
History length k	1	4
Target	max (standard)	Double DQN
Output head	linear	dueling ($V + \text{advantage}$)
Hidden layers		2×128 , ReLU
Optimiser, learning rate		Adam, 5×10^{-4}
Discount γ_{RL}		0.99
Replay capacity, warm-up		50 000, 1 000 transitions
Minibatch size		64
Target update period C		1 000 steps (hard copy)
Loss, gradient clip		Huber ($\delta = 1$), $\ \nabla\ _2 \leq 10$
Exploration	ϵ : $1.0 \rightarrow 0.01$, exponential (0.05 at 50% of budget)	
Action set	{EASY, HARD, EDF, SPT, BREAK}	
Training budget	5 000 episodes = 240 000 decisions	
Checkpoint selection	best greedy return on 100 validation days, every 100 episodes	

Table 5: Performance on 500 held-out test days (nominal environment). Learned policies: mean $\pm$ s.d. across ten training seeds of the per-seed test means. Heuristics: mean $\pm$ s.d. across test days (they are deterministic given the day). Bold marks the best value in each column.

Method	Return	On-time rate	Weighted on-time	Mean attention	Mean fatigue	Break share
Random	8.85 $\pm$ 2.71	0.535 $\pm$ 0.131	0.488 $\pm$ 0.143	0.855 $\pm$ 0.037	0.207 $\pm$ 0.045	0.326 $\pm$ 0.065
FIFO	3.49 $\pm$ 3.49	0.432 $\pm$ 0.139	0.422 $\pm$ 0.125	0.463 $\pm$ 0.106	0.546 $\pm$ 0.045	0.000 $\pm$ 0.003
EDF	4.24 $\pm$ 3.82	0.468 $\pm$ 0.155	0.454 $\pm$ 0.137	0.465 $\pm$ 0.105	0.545 $\pm$ 0.046	0.000 $\pm$ 0.003
SPT	7.95 $\pm$ 2.94	0.625 $\pm$ 0.120	0.494 $\pm$ 0.122	0.543 $\pm$ 0.093	0.477 $\pm$ 0.038	0.000 $\pm$ 0.001
EDF-Periodic	11.41 $\pm$ 3.32	0.660 $\pm$ 0.160	0.657 $\pm$ 0.148	0.798 $\pm$ 0.040	0.296 $\pm$ 0.029	0.188 $\pm$ 0.005
EDF-Threshold	11.35 $\pm$ 3.28	0.654 $\pm$ 0.164	0.650 $\pm$ 0.155	0.806 $\pm$ 0.022	0.288 $\pm$ 0.034	0.201 $\pm$ 0.037
SPT-Periodic	12.25 $\pm$ 2.04	0.709 $\pm$ 0.096	0.591 $\pm$ 0.111	0.817 $\pm$ 0.034	0.261 $\pm$ 0.023	0.188 $\pm$ 0.005
SPT-Threshold	12.30 $\pm$ 2.03	0.714 $\pm$ 0.099	0.597 $\pm$ 0.114	0.813 $\pm$ 0.022	0.268 $\pm$ 0.036	0.187 $\pm$ 0.035
Cog-Heuristic	10.59 $\pm$ 2.76	0.638 $\pm$ 0.132	0.533 $\pm$ 0.150	0.803 $\pm$ 0.034	0.279 $\pm$ 0.020	0.198 $\pm$ 0.025
DQN	10.29 $\pm$ 0.26	0.677 $\pm$ 0.008	0.579 $\pm$ 0.012	0.677 $\pm$ 0.017	0.394 $\pm$ 0.018	0.108 $\pm$ 0.020
AA-DQN (ours)	12.17 $\pm$ 0.15	0.728 $\pm$ 0.005	0.655 $\pm$ 0.012	0.754 $\pm$ 0.011	0.333 $\pm$ 0.012	0.150 $\pm$ 0.018

Table 6: Paired comparison of AA-DQN with each baseline on 500 common-random-number test days: mean difference with 95% bootstrap CI (on-time rates in percentage points), matched-pairs rank-biserial correlation r_{rb} (positive favours AA-DQN), and Holm-adjusted Wilcoxon signed-rank p -values.

AA-DQN vs.	Episode return			On-time rate (pp)			Weighted on-time rate (pp)		
	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}
Random	+3.32 [+3.13, +3.52]	0.97	$< 10^{-4}$	+19.3 [+18.3, +20.3]	1.00	$< 10^{-4}$	+16.7 [+15.8, +17.6]	0.98	$< 10^{-4}$
FIFO	+8.68 [+8.47, +8.90]	1.00	$< 10^{-4}$	+29.6 [+28.7, +30.5]	1.00	$< 10^{-4}$	+23.3 [+22.6, +24.0]	1.00	$< 10^{-4}$
EDF	+7.93 [+7.71, +8.17]	1.00	$< 10^{-4}$	+26.0 [+25.1, +26.9]	1.00	$< 10^{-4}$	+20.0 [+19.3, +20.7]	1.00	$< 10^{-4}$
SPT	+4.22 [+4.06, +4.39]	1.00	$< 10^{-4}$	+10.2 [+9.5, +10.9]	0.96	$< 10^{-4}$	+16.1 [+15.3, +16.8]	1.00	$< 10^{-4}$
EDF-Periodic	+0.76 [+0.56, +0.94]	0.34	$< 10^{-4}$	+6.7 [+5.9, +7.6]	0.69	$< 10^{-4}$	-0.2 [-0.9, +0.5]	-0.05	0.6403
EDF-Threshold	+0.82 [+0.64, +1.00]	0.36	$< 10^{-4}$	+7.4 [+6.5, +8.3]	0.72	$< 10^{-4}$	+0.5 [-0.3, +1.2]	0.01	0.8269
SPT-Periodic	-0.08 [-0.24, +0.08]	-0.06	0.2860	+1.9 [+1.1, +2.6]	0.24	$< 10^{-4}$	+6.3 [+5.5, +7.2]	0.62	$< 10^{-4}$
SPT-Threshold	-0.13 [-0.29, +0.02]	-0.08	0.2390	+1.4 [+0.7, +2.1]	0.20	0.0002	+5.7 [+4.9, +6.6]	0.61	$< 10^{-4}$
Cog-Heuristic	+1.58 [+1.43, +1.72]	0.80	$< 10^{-4}$	+8.9 [+8.3, +9.6]	0.92	$< 10^{-4}$	+12.2 [+11.4, +13.1]	0.93	$< 10^{-4}$